

GED0081 — College Physics 1

FINAL EXAM REVIEWER

Modules 1–9 · Vectors through Rotational Equilibrium & Torque

EXAM DETAILS

Saturday, July 18 · 9:00–11:00 AM
Room 303 (DW21) — arrive early
50 items total, Modules 1–9

HOW TO USE THIS

1. Read the Quick-Reference Formula Sheet first.
2. Study each module's key ideas + worked example.
3. Cover the answers, attempt every practice item.

Exam Item Breakdown

Use this to weight your study time — modules with more items deserve more of your remaining hours.

Module	Topic	Items
M1	Introduction to Vectors & Unit Vectors	7
M2	Operations on Vectors (add, dot, cross)	5
M3	Motion in 1D (kinematics)	8
M4	Motion in 2D (projectile & circular motion)	4
M5	Newton's Laws of Motion (concepts & FBDs)	8
M6	Applications of Newton's Laws (friction, inclines, pulleys)	6
M7	Work, Energy & Power	6
M8	Impulse & Momentum	3
M9	Equilibrium & Torque	3
TOTAL		50

What's most worth your remaining hours

- M3 and M5 tie for the most items (8 each) — 1D kinematics equations and Newton's Law concepts/FBDs deserve top priority.
- M6 (6 items) is the most calculation-heavy module — friction, inclines, and pulley systems. Budget extra practice time here.
- M7 (6 items) mixes short concept questions (scalar/vector, definitions) with a couple of computations — cheap points if you know the formulas.
- M8 and M9 are only 3 items each — know the core formula and one worked pattern per module; don't over-invest here.

Contents

M1 Introduction to Vectors *7 items*

M2 Operations on Vectors *5 items*

M3 Motion in 1D *8 items*

M4 Motion in 2D *4 items*

M5 Newton's Laws of Motion *8 items*

M6 Applications of Newton's Laws *6 items*

M7 Work, Energy & Power *6 items*

M8 Impulse & Momentum *3 items*

M9 Equilibrium & Torque *3 items*

Quick-Reference Formula Sheet

Final Countdown Checklist

Quick-Reference Formula Sheet

Everything on one map. If you only have 20 minutes before the exam, review this page.

M1–M2 · Vectors

Quantity	Formula / Notes
Magnitude (2D)	$ A = \sqrt{A_x^2 + A_y^2}$
Magnitude (3D)	$ A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
Angle of inclination	$\theta = \tan^{-1}(A_y / A_x)$, measured from +x axis
Unit vector	$u = A / A $
Dot product	$A \cdot B = A B \cos \theta = A_x B_x + A_y B_y + A_z B_z$ (result is a SCALAR)
Cross product	$A \times B = (A_y B_z - A_z B_y)\hat{i} - (A_x B_z - A_z B_x)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$ (result is a VECTOR)

M3 · Motion in 1D

Quantity	Formula / Notes
Average speed	total distance / total time
Average velocity	displacement / elapsed time
Kinematics Eq. 1	$v_f = v_0 + at$
Kinematics Eq. 2	$\Delta x = ((v_0 + v_f)/2) t$
Kinematics Eq. 3	$\Delta x = v_0 t + \frac{1}{2}at^2$
Kinematics Eq. 4	$v_f^2 = v_0^2 + 2a\Delta x$
Free fall	same 4 equations, $a = g = -9.8 \text{ m/s}^2$ (down is negative)

M4 · Motion in 2D

Quantity	Formula / Notes
Horizontal displacement	$x = v_0 \cos \theta \cdot t$
Vertical displacement	$y = v_0 \sin \theta \cdot t - \frac{1}{2}gt^2$
Time to peak	$t = v_0 \sin \theta / g$
Total flight time	$t' = 2v_0 \sin \theta / g$
Max height	$H = (v_0 \sin \theta)^2 / 2g$
Range	$R = v_0^2 \sin 2\theta / g$ (max range at $\theta = 45^\circ$)
Centripetal accel.	$a_c = v^2 / r = 4\pi^2 r / t^2$
Centripetal force	$F_c = m a_c = mv^2 / r$
Frequency ↔ period	$f = 1/t$ (Hz); $v = 2\pi r / t = 2\pi r f$

M5–M6 · Newton's Laws & Applications

Quantity	Formula / Notes
Newton's 2nd Law	$F = ma \rightarrow a = F/m \rightarrow m = F/a$
Weight	$W = mg$ (always points down)
Incline, x-component	$W_x = W \sin \theta$ (down the slope)
Incline, y-component / Normal	$W_y = N = W \cos \theta$ (frictionless, no y-motion)
Friction force	$f = \mu N$ ($\mu_s > \mu_k$ always)
Incline w/ friction, sliding down	$a = g \sin \theta - \mu g \cos \theta$
Pulley: block + hanging mass	$a = m_2g / (m_1 + m_2)$; $T = m_1a$
Atwood machine ($m_1 > m_2$)	$a = (m_1 - m_2)g / (m_1 + m_2)$; $T = m_2(g + a)$
Elevator tension (moving up)	$T = m(g + a)$
Elevator tension (moving down/decel.)	$T = m(g - a)$

M7 · Work, Energy & Power

Quantity	Formula / Notes
Work	$W = Fd \cos \theta$ (SCALAR; unit = Joule, J)
Net work / work-energy theorem	$W_{\text{net}} = KE_f - KE_0 = \Delta KE$
Kinetic energy	$KE = \frac{1}{2}mv^2$
Potential energy (gravitational)	$PE = mgh$
Power	$P = W/t$ (unit = Watt; 1 W = 1 J/s); also $P = \text{Force} \times \text{velocity}$

M8 · Impulse & Momentum

Quantity	Formula / Notes
Momentum	$p = mv$
Impulse	$I = \Delta p = m(v_f - v_0)$ (unit: $N \cdot s = kg \cdot m/s$)
Conservation (elastic)	$m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$; KE conserved
Perfectly inelastic (objects stick)	$m_1v_1 + m_2v_2 = (m_1 + m_2)v_f$; KE NOT conserved

M9 · Equilibrium & Torque

Quantity	Formula / Notes
1st condition of equilibrium	$\Sigma F = 0$ ($\Sigma x\text{-forces} = 0$ and $\Sigma y\text{-forces} = 0$ separately)
2nd condition of equilibrium	$\Sigma \text{Torque} = 0$ ($\Sigma \text{clockwise} = \Sigma \text{counterclockwise}$)
Torque	$\tau = rF \sin \theta$ (unit: $N \cdot m$). Max when $\theta = 90^\circ$; zero when force is parallel to r

M1 · Introduction to Vectors

7 items on the exam

Key Ideas

- A scalar has magnitude only (time, mass, temperature, density). A vector has both magnitude AND direction (displacement, velocity, force).
- Two vectors are equivalent if same magnitude; parallel if same direction; anti-parallel if opposite directions; equal if same magnitude AND direction.
- Unit vectors $\hat{i}=(1,0,0)$, $\hat{j}=(0,1,0)$, $\hat{k}=(0,0,1)$ each have magnitude 1 and point along +x, +y, +z respectively.
- Any vector can be written as $A = A_x \hat{i} + A_y \hat{j}$ (2D) or $A = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ (3D).
- To find which quadrant/plane/axis a vector sits on, look at the signs of its components. Positive angle = measured counterclockwise from +x axis; negative angle = clockwise.

Core Formulas

$$|A| = \sqrt{A_x^2 + A_y^2} \quad [2D \text{ magnitude}]$$

$$|A| = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad [3D \text{ magnitude}]$$

$$\theta = \tan^{-1}(A_y / A_x) \quad [\text{angle from +x axis}]$$

$$u = A / |A| \quad [\text{unit vector in direction of } A]$$

Worked Example — Magnitude, angle, and unit vector

Given: Vector $F = \langle 6, -8 \rangle$

Required: $|F|$, θ , and the unit vector u

- Magnitude: $|F| = \sqrt{6^2 + (-8)^2} = \sqrt{36+64} = \sqrt{100} = 10$
- Angle: $\theta = \tan^{-1}(-8/6) = \tan^{-1}(-1.333) = -53.13^\circ$ (below +x axis, since A_y is negative and A_x is positive $\rightarrow$ Quadrant IV)
- Unit vector: $u = \langle 6, -8 \rangle / 10 = \langle 0.6, -0.8 \rangle$

Answer: $|F| = 10$, $\theta = -53.13^\circ$, $u = \langle 0.6, -0.8 \rangle$

Practice Questions

1 [M1] Classify: is 'acceleration' a scalar or a vector quantity? Why?

Answer: Vector — it has both a magnitude (how fast velocity is changing) and a direction (the direction of that change).

2 [M1] Vector $B = 5\hat{i} - 5\hat{j} + 0\hat{k}$. Which plane does it lie in?

Answer: The xy-plane, since its z-component is 0.

3 [M1] Find the magnitude of vector $V = \langle 4, -3, 12 \rangle$.

Answer: $|V| = \sqrt{4^2 + (-3)^2 + 12^2} = \sqrt{16+9+144} = \sqrt{169}$

$|V| = 13$

4 [M1] Find the angle of inclination of the vector $\langle -7, 7 \rangle$ (give the angle from the +x axis, and state the quadrant).

Answer: $\theta = \tan^{-1}(7/-7) = \tan^{-1}(-1) \rightarrow$ reference angle 45° . Since A_x is negative and A_y is positive, the vector is in Quadrant II, so $\theta = 180^\circ - 45^\circ$

$\theta = 135^\circ$

5 [M1] Find the unit vector in the same direction as $A = 3\hat{i} + 4\hat{j}$.

Answer: $|A| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

$u = \langle 3/5, 4/5 \rangle = \langle 0.6, 0.8 \rangle$

6 [M1] A vector is given as 8 m, 40° S of E. What does 'S of E' tell you about which quadrant it's in?

Answer: 'South of East' means you start facing East and rotate toward South — this places the vector in Quadrant IV (positive x, negative y).

7 [M1] True or False: If two vectors have the same magnitude but point in opposite directions, they are considered equal.

Answer: False — they are anti-parallel, not equal. 'Equal' requires both the same magnitude AND the same direction.

M2 · Operations on Vectors

5 items on the exam

Key Ideas

- Vector addition (graphical): place vectors tip-to-tail; the resultant runs from the first tail to the last tip.
- Component method: add the x-components together and the y-components together separately, then recombine: $R_x = \sum F_x$, $R_y = \sum F_y$, and $|R| = \sqrt{R_x^2 + R_y^2}$.
- The equilibrant is the vector that exactly cancels the resultant — same magnitude, opposite direction.
- Dot product (scalar/inner product) → returns a NUMBER, tells you about the angle between two vectors. $A \cdot B = 0$ means the vectors are perpendicular.
- Cross product → returns a VECTOR perpendicular to both original vectors (right-hand rule gives its direction). Used to compute torque, among other things.

Core Formulas

$$A \cdot B = |A||B| \cos \theta = A_x B_x + A_y B_y + A_z B_z$$

$$A \times B = (A_y B_z - A_z B_y)\hat{i} - (A_x B_z - A_z B_x)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

Worked Example — Dot product from components

Given: $A = 3\hat{i} - \hat{j} + 2\hat{k}$, $B = -\hat{i} + 4\hat{j} - 2\hat{k}$

Required: $A \cdot B$

- $A \cdot B = A_x B_x + A_y B_y + A_z B_z$
- $A \cdot B = (3)(-1) + (-1)(4) + (2)(-2)$
- $A \cdot B = -3 + (-4) + (-4) = -11$

Answer: $A \cdot B = -11$

Worked Example — Resultant of two forces (component method)

Given: $F_1 = 40 \text{ N}$ at 0° (along +x); $F_2 = 30 \text{ N}$ at 90° (along +y)

Required: Magnitude and direction of the resultant

- $R_x = 40 + 0 = 40 \text{ N}$; $R_y = 0 + 30 = 30 \text{ N}$
- $|R| = \sqrt{40^2 + 30^2} = \sqrt{1600 + 900} = \sqrt{2500} = 50 \text{ N}$
- $\theta = \tan^{-1}(30/40) = \tan^{-1}(0.75) = 36.87^\circ$ above +x axis

Answer: $R = 50 \text{ N}$ at 36.87° from +x axis

Practice Questions

8

[M2] What is the dot product of two vectors that are perpendicular to each other?

Answer: Zero — since $\cos 90^\circ = 0$, $A \cdot B = |A||B|\cos 90^\circ = 0$ regardless of magnitude.

9

[M2] What is the dot product of two identical (coinciding) vectors of magnitude 6?

Answer: $A \cdot B = |A||B|\cos 0^\circ = (6)(6)(1) = 36$ — dot product is maximized when vectors point the same way.

10

[M2] Find $A \cdot B$ given $|A| = 9$, $|B| = 5$, angle between them = 60° .

Answer: $A \cdot B = |A||B|\cos \theta = (9)(5)(\cos 60^\circ) = 45(0.5)$

$A \cdot B = 22.5$

11

[M2] Two ropes pull on a ring: Rope 1 pulls 25 N due East, Rope 2 pulls 25 N due North. Find the magnitude of the resultant force.

Answer: $R_x = 25 \text{ N}$, $R_y = 25 \text{ N}$

$$|R| = \sqrt{(25^2 + 25^2)} = \sqrt{1250} = 35.36 \text{ N}$$

12

[M2] Given $P = 2\hat{i} + \hat{j}$ and $Q = \hat{i} - 3\hat{j}$, find $P \times Q$. (Treat both as lying in the xy-plane, so z-components are 0.)

Answer: $P \times Q = (P_y Q_z - P_z Q_y)\hat{i} - (P_x Q_z - P_z Q_x)\hat{j} + (P_x Q_y - P_y Q_x)\hat{k}$

Since $P_z = Q_z = 0$: only the $\hat{k}$ term survives $\rightarrow (P_x Q_y - P_y Q_x)\hat{k} = [(2)(-3) - (1)(1)]\hat{k} = (-6-1)\hat{k}$

$$P \times Q = -7\hat{k}$$

M3 · Motion in 1D

8 items on the exam — the single highest-weighted module, tied with M5. Know this cold.

Key Ideas

- Distance is a scalar (total path length, always positive). Displacement is a vector (net change in position, $x_f - x_i$; can be negative).
- Speed = total distance / total time (scalar). Velocity = displacement / elapsed time (vector, can be negative).
- On a position-vs-time graph: the SLOPE is the velocity. Zero slope = at rest. On a velocity-vs-time graph: the SLOPE is the acceleration.
- An object is accelerating whenever its velocity is changing — this includes speeding up, slowing down, OR simply changing direction (like in circular motion).
- Free fall uses the same 4 kinematics equations, with $a = g = 9.8 \text{ m/s}^2$ downward. Standard convention: up is positive, down is negative, so $g = -9.8 \text{ m/s}^2$.
- At the peak of an object thrown straight up, $v_f = 0$ (this is the key trick for 'how high' and 'how long to reach the top' problems).

The 4 Kinematics Equations — know when to use each

Equation	Use when you don't have / aren't asked for...
$v_f = v_0 + at$	... Δx (position/displacement)
$\Delta x = ((v_0 + v_f)/2)t$	...acceleration, a
$\Delta x = v_0t + \frac{1}{2}at^2$	...final velocity, v_f
$v_f^2 = v_0^2 + 2a\Delta x$	...time, t

Note: Pick the equation that already contains BOTH your given values and your required value — that's the fastest path, no need to solve for an intermediate variable first.

Worked Example — Braking distance

Given: $v_0 = 25 \text{ m/s}$, $v_f = 0 \text{ m/s}$, $t = 4 \text{ s}$

Required: $a = ?$ and $\Delta x = ?$

- a : use $v_f = v_0 + at \rightarrow 0 = 25 + a(4) \rightarrow a = -25/4$
- $a = -6.25 \text{ m/s}^2$
- Δx : use $\Delta x = ((v_0 + v_f)/2)t = ((25+0)/2)(4) = (12.5)(4)$

Answer: $a = -6.25 \text{ m/s}^2$, $\Delta x = 50 \text{ m}$

Worked Example — Free fall — thrown upward

Given: $v_0 = 18 \text{ m/s}$ (up), $g = -9.8 \text{ m/s}^2$, at the peak $v_f = 0$

Required: Maximum height H and total time of flight t'

- H : use $v_f^2 = v_0^2 + 2g\Delta y \rightarrow 0 = 18^2 + 2(-9.8)\Delta y \rightarrow \Delta y = -324 / -19.6$
- $H = 16.53 \text{ m}$
- Time to peak: $v_f = v_0 + gt \rightarrow 0 = 18 + (-9.8)t \rightarrow t = 1.84 \text{ s}$
- Total flight time (up and back down to same height) $t' = 2t = 3.67 \text{ s}$

Answer: $H = 16.53 \text{ m}$, $t' = 3.67 \text{ s}$

Practice Questions

13

[M3] A runner jogs 300 m east, then turns around and jogs 100 m west, all in 80 seconds. Find (a) the distance and (b) the displacement.

Answer: (a) Distance = $300 + 100 = 400$ m

(b) Displacement = $300 - 100 = 200$ m east

14

[M3] Using the runner from the previous question, find the average speed and average velocity.

Answer: Average speed = total distance / total time = $400 \text{ m} / 80 \text{ s} = 5 \text{ m/s}$

Average velocity = displacement / elapsed time = $200 \text{ m} / 80 \text{ s} = 2.5 \text{ m/s east}$

15

[M3] A car accelerates from 10 m/s to 34 m/s in 6 seconds. Find its acceleration.

Answer: $v_f = v_0 + at \rightarrow 34 = 10 + a(6) \rightarrow a = 24/6$

$a = 4 \text{ m/s}^2$

16

[M3] A ball starts from rest and accelerates at 5 m/s^2 for 3 seconds. How far does it travel?

Answer: $\Delta x = v_0 t + \frac{1}{2}at^2 = (0)(3) + \frac{1}{2}(5)(3^2) = 0 + \frac{1}{2}(5)(9)$

$\Delta x = 22.5 \text{ m}$

17

[M3] On a position-vs-time graph, a line is perfectly horizontal (flat) from $t=2\text{s}$ to $t=6\text{s}$. What does this tell you about the object's motion?

Answer: The slope is zero, so velocity = 0 during that interval — the object is at rest (not moving) from $t=2\text{s}$ to $t=6\text{s}$.

18

[M3] A stone is dropped ($v_0 = 0$) from a bridge and hits the water 3.2 s later. How high is the bridge, and how fast is the stone moving when it hits the water?

Answer: Height: $\Delta y = v_0 t + \frac{1}{2}gt^2 = 0 + \frac{1}{2}(-9.8)(3.2^2) = \frac{1}{2}(-9.8)(10.24)$

$\Delta y = -50.18 \text{ m} \rightarrow$ bridge is 50.18 m high

Speed: $v_f = v_0 + gt = 0 + (-9.8)(3.2) = -31.36 \text{ m/s}$

Height = 50.18 m, $v_f = 31.36 \text{ m/s downward}$

19

[M3] A ball is thrown straight up and returns to the thrower's hand after 5 seconds. What was its initial velocity?

Answer: Total flight time $t' = 5 \text{ s} \rightarrow$ time to reach peak $t = t'/2 = 2.5 \text{ s}$

At the peak, $v_f = 0$. Use $v_f = v_0 + gt \rightarrow 0 = v_0 + (-9.8)(2.5) \rightarrow v_0 = 24.5$

$v_0 = 24.5 \text{ m/s}$

20

[M3] Two cars start from the same point. Car A moves at constant 20 m/s. Car B starts from rest and accelerates at 4 m/s^2 . After how many seconds will Car B catch up to Car A? (Hint: set their displacements equal.)

Answer: Car A's position: $x_A = 20t$. Car B's position: $x_B = \frac{1}{2}(4)t^2 = 2t^2$

Set equal: $20t = 2t^2 \rightarrow 20 = 2t$ (dividing both sides by t , $t \neq 0$) $\rightarrow t = 10$

$t = 10 \text{ seconds}$

M4 · Motion in 2D

4 items on the exam — covers projectile motion AND circular motion

Key Ideas — Projectile Motion

- Projectile motion is two INDEPENDENT motions happening at once: constant velocity horizontally ($a_x=0$), and free-fall (uniform acceleration) vertically ($a_y = -g$).
- At the very top of the trajectory (the peak), the vertical velocity component $v_y = 0$ — but the horizontal velocity v_x is NOT zero at the peak (it stays constant the whole flight).
- Symmetrical path (launched and lands at same height): use the angled-launch formulas directly. Non-symmetrical (e.g. rolls off a table, or launched from a cliff): treat horizontal and vertical separately.
- Maximum range occurs when the launch angle $\theta = 45^\circ$.

Key Ideas — Uniform Circular Motion

- Centripetal acceleration always points TOWARD the center of the circle, even though the object moves at constant speed — because direction is constantly changing, the object IS accelerating.
- Centripetal force is what causes this — it's the net inward force (could be tension, gravity, friction, normal force, etc. depending on the situation), not a separate new kind of force.
- 1 revolution = $360^\circ = 2\pi$ radians. Always convert to radians before using $s = r\theta$.

Core Formulas

Symmetrical launch — Time to peak: $t = v_0 \sin\theta / g$

Total flight time: $t' = 2v_0 \sin\theta / g$

Max height: $H = (v_0 \sin\theta)^2 / 2g$

Range: $R = v_0^2 \sin(2\theta) / g$

Centripetal acceleration: $a_c = v^2 / r = 4\pi^2 r / t^2$

Centripetal force: $F_c = ma_c = mv^2 / r$

Worked Example — Projectile launched at an angle

Given: $v_0 = 24$ m/s at $\theta = 40^\circ$ above horizontal

Required: Time of flight t' , max height H , and range R

- $t' = 2v_0 \sin\theta / g = 2(24)(\sin 40^\circ) / 9.8 = 2(24)(0.643)/9.8$
- $t' = 3.15$ s
- $H = (v_0 \sin\theta)^2 / 2g = (24 \times 0.643)^2 / (2 \times 9.8) = (15.43)^2 / 19.6$
- $H = 12.15$ m
- $R = v_0^2 \sin(2\theta) / g = (24^2)(\sin 80^\circ) / 9.8 = (576)(0.985)/9.8$

Answer: $t' = 3.15$ s, $H = 12.15$ m, $R = 57.87$ m

Worked Example — Centripetal acceleration and force

Given: $r = 50$ m, $v = 15$ m/s, $m = 900$ kg (a car rounding a curve)

Required: $a_c = ?$ and $F_c = ?$

- $a_c = v^2/r = 15^2 / 50 = 225/50$
- $a_c = 4.5$ m/s²
- $F_c = ma_c = (900)(4.5)$

Answer: $a_c = 4.5 \text{ m/s}^2$, $F_c = 4050 \text{ N}$

Practice Questions

21

[M4] A stone rolls off a horizontal table 2.0 m high and lands 3.5 m away from the base. Find (a) the time of flight and (b) the initial horizontal velocity.

Answer: (a) Since $v_{0y} = 0$ (rolls off horizontally), use $\Delta y = \frac{1}{2}gt^2 \rightarrow 2.0 = \frac{1}{2}(9.8)t^2 \rightarrow t^2 = 4/9.8 \rightarrow t = 0.639 \text{ s}$

(b) $R = v_{0x} \cdot t \rightarrow 3.5 = v_{0x}(0.639) \rightarrow v_{0x} = 5.48$

$t = 0.639 \text{ s}$, $v_{0x} = 5.48 \text{ m/s}$

22

[M4] A ball is kicked at 20 m/s at 35° above the horizontal. What is its maximum height?

Answer: $H = (v_0 \sin \theta)^2 / 2g = (20 \times \sin 35^\circ)^2 / (2 \times 9.8) = (20 \times 0.574)^2 / 19.6 = (11.47)^2 / 19.6$

$H = 6.71 \text{ m}$

23

[M4] At what launch angle is the horizontal range of a projectile maximized (for a fixed launch speed and no air resistance)?

Answer: 45° — this angle gives the best balance between horizontal speed and time in the air.

24

[M4] A ball on a string moves in a horizontal circle of radius 0.8 m, completing one revolution every 2 seconds. Find its centripetal acceleration.

Answer: Use $a_c = 4\pi^2 r / t^2$ (t = time for 1 revolution) = $4\pi^2(0.8) / 2^2$

$a_c = 4\pi^2(0.8)/4 = \pi^2(0.8) = 7.895$

$a_c = 7.90 \text{ m/s}^2$

M5 · Newton's Laws of Motion

8 items on the exam — tied with M3 for the most items. Concept-heavy: expect definitional and FBD-reasoning questions, not just plug-and-chug.

Key Ideas — The Three Laws

- 1st Law (Inertia): An object at rest stays at rest, and an object in motion stays in motion (same speed, same direction), UNLESS acted on by a net (unbalanced) external force.
- 2nd Law: $F = ma$. Acceleration is directly proportional to net force, and inversely proportional to mass.
- 3rd Law: For every action force, there is an equal and opposite reaction force. These two forces act on DIFFERENT objects, so they never cancel each other out.

Types of Forces (mnemonic: W.A.N.T. Food)

Force	Notes
Weight (W)	$W = mg$. Always points straight down (toward Earth). Present whenever an object has mass.
Applied Force	Any push or pull described directly in the problem.
Normal Force (N)	Always perpendicular to the contact surface. Present whenever the object touches a surface.
Tension (T)	Present whenever a rope/string/cable is attached. # of strings = # of tension forces.
Friction (f)	$f = \mu N$. Opposes motion (or impending motion). Zero if the surface is frictionless/smooth.

Free-Body Diagram (FBD) Checklist

- Draw ONLY the forces acting ON the object — never the forces the object exerts on its environment (no both halves of an action-reaction pair on the same diagram).
- If forces are at an angle, break them into perpendicular x- and y-components before summing.
- $\Sigma F_x = \max$ and $\Sigma F_y = \max$ are solved SEPARATELY. If there's no motion along an axis, that axis's acceleration = 0.

Worked Example — Basic $F=ma$

Given: A 12 kg crate experiences a net force of 18 N

Required: acceleration a

- $a = F/m = 18/12$

Answer: $a = 1.5 \text{ m/s}^2$

Worked Example — Conceptual — comparing accelerations

Given: Two forces of equal magnitude (20 N) are applied to two different objects: Object A has mass 4 kg, Object B has mass 10 kg

Required: Which object has the greater acceleration, and by what factor?

- $a_A = F/m_A = 20/4 = 5 \text{ m/s}^2$
- $a_B = F/m_B = 20/10 = 2 \text{ m/s}^2$
- Ratio: $a_A/a_B = 5/2 = 2.5$

Answer: Object A accelerates 2.5× faster than Object B (mass and acceleration are inversely related when force is constant)

Practice Questions

25

[M5] A book sits still on a table. According to Newton's 1st Law, what does this tell you about the net force acting on it?

Answer: The net force on the book must be zero — the weight (down) and the normal force (up) are perfectly balanced, so it stays at rest.

26

[M5] A 5 kg object accelerates at 3 m/s^2 when a force F is applied. If the SAME force is applied to a 15 kg object, what is the new acceleration?

Answer: First find F : $F = ma = (5)(3) = 15 \text{ N}$

Now solve for the new acceleration: $a = F/m = 15/15$

$a = 1 \text{ m/s}^2$

27

[M5] You are a passenger in a car that suddenly brakes hard. Your body lurches forward. Which of Newton's Laws explains this, and why?

Answer: The 1st Law (Inertia) — your body was in motion along with the car, and tends to stay in motion at the same speed even after the car itself slows down, until the seatbelt (an external force) acts on you.

28

[M5] A book rests on a frictionless table. List every force acting on the book.

Answer: Only two: Weight (W , pointing down) and Normal force (N , pointing up). No applied force is mentioned, no string is attached, and the table is frictionless.

29

[M5] When you push against a wall, the wall pushes back on you with equal force. Which law is this, and why don't the two forces cancel out?

Answer: Newton's 3rd Law. The two forces don't cancel because they act on two DIFFERENT objects — one acts on the wall, the other acts on you. Forces only cancel when they act on the SAME object.

30

[M5] If the net force on a moving object is exactly zero, what happens to its velocity?

Answer: Nothing changes — the object continues moving at constant velocity (same speed, same direction). Zero net force means zero acceleration, not zero velocity.

31

[M5] A 4 kg box is pulled by a rope with an applied force of 20 N. Find the box's acceleration, assuming the surface is frictionless.

Answer: Only the applied force acts along the direction of motion: $\Sigma F_x = F_{\text{app}} = \text{max} \rightarrow 20 = 4a \rightarrow a = 20/4$

$a = 5 \text{ m/s}^2$

32

[M5] Why must weight always be included in a free-body diagram, but tension is only sometimes included?

Answer: Weight ($W = mg$) exists for ANY object with mass, so it's always present. Tension only exists if the object is attached to a rope, string, or cable — if there's no string in the problem, there's no tension force to draw.

M6 · Applications of Newton's Laws

6 items on the exam — the most computation-heavy module. Friction, inclines, and connected systems (pulleys).

Key Ideas

- On an incline, tilt your x-y axes to match the slope: x-axis runs along the surface, y-axis is perpendicular to it. Then $W_x = W \sin\theta$ (pulls the object down the slope) and $W_y = W \cos\theta$ (presses into the surface, balanced by the Normal force).
- Static friction (μ_s) resists the START of motion; kinetic friction (μ_k) resists motion that's ALREADY happening. μ_s is always greater than μ_k .
- When two objects are in contact and pushed together (like Box A pushing Box B), they share the SAME acceleration — but the force each one individually experiences differs based on its own mass.
- For pulley systems, first determine which way the system will actually move by comparing the forces — the direction of the LARGER pulling force wins.
- Elevator/apparent weight problems: moving up & speeding up (or down & slowing down) → tension is GREATER than weight. Moving down & speeding up (or up & slowing down) → tension is LESS than weight.

Core Formulas

Friction: $f = \mu N$

Incline, frictionless, sliding down: $a = g \sin\theta$

Incline, with friction, sliding down: $a = g \sin\theta - \mu g \cos\theta$

Normal force on an incline: $N = mg \cos\theta$

Block (m_1 , on table) + hanging mass (m_2) over pulley: $a = m_2g/(m_1+m_2)$; $T = m_1a$

Atwood machine ($m_1 > m_2$, both hanging): $a = (m_1 - m_2)g/(m_1 + m_2)$; $T = m_2(g + a)$

Worked Example — Block sliding down a rough incline

Given: $m = 10 \text{ kg}$, $\theta = 25^\circ$, $\mu_k = 0.15$

Required: acceleration down the incline

- $a = g \sin\theta - \mu g \cos\theta$
- $a = (9.8)(\sin 25^\circ) - (0.15)(9.8)(\cos 25^\circ)$
- $a = (9.8)(0.4226) - (0.15)(9.8)(0.9063)$
- $a = 4.14 - 1.332$

Answer: $a = 2.81 \text{ m/s}^2$ down the incline

Worked Example — Block on table connected to hanging mass

Given: $m_1 = 6 \text{ kg}$ (on frictionless table), $m_2 = 4 \text{ kg}$ (hanging), connected over a pulley

Required: acceleration a and tension T

- $a = m_2g / (m_1 + m_2) = (4)(9.8) / (6 + 4) = 39.2/10$
- $a = 3.92 \text{ m/s}^2$
- $T = m_1a = (6)(3.92)$

Answer: $a = 3.92 \text{ m/s}^2$, $T = 23.52 \text{ N}$

Practice Questions

33 [M6] A 15 kg block rests on a 20° frictionless incline. Find the normal force acting on it.

Answer: $N = mg \cos\theta = (15)(9.8)(\cos 20^\circ) = (147)(0.9397)$

$N = 138.13 \text{ N}$

34 [M6] Using the same 15 kg block on a 20° frictionless incline, find its acceleration as it slides down.

Answer: $a = g \sin\theta = (9.8)(\sin 20^\circ) = (9.8)(0.342)$

$a = 3.35 \text{ m/s}^2$ down the incline

35 [M6] A wooden crate ($m = 8 \text{ kg}$) sits on a rough horizontal floor. The coefficient of kinetic friction is 0.3. Find the friction force needed to overcome once the crate starts sliding.

Answer: On a horizontal surface, $N = W = mg = (8)(9.8) = 78.4 \text{ N}$

$f = \mu_k N = (0.3)(78.4)$

$f = 23.52 \text{ N}$

36 [M6] Two blocks are pushed together along a frictionless floor: Box A (3 kg) is behind Box B (7 kg), pushed by a total applied force of 40 N. Find the acceleration of the system and the force Box A exerts on Box B.

Answer: System acceleration: $a = F_{\text{total}} / m_{\text{total}} = 40 / (3+7) = 40/10 = 4 \text{ m/s}^2$

Since Box A pushes Box B, the contact force equals the force needed to accelerate Box B alone: $F(A \text{ on } B) = m_B \times a = (7)(4)$

$a = 4 \text{ m/s}^2$, Force on Box B = 28 N

37 [M6] An elevator (mass 500 kg) accelerates upward at 1.5 m/s^2 . Find the tension in the supporting cable.

Answer: $T = m(g+a) = 500(9.8 + 1.5) = 500(11.3)$

$T = 5650 \text{ N}$

38 [M6] An Atwood machine has $m_1 = 8 \text{ kg}$ and $m_2 = 5 \text{ kg}$ hanging on either side of a pulley. Find the system's acceleration and the tension in the rope.

Answer: $a = (m_1 - m_2)g / (m_1 + m_2) = (8-5)(9.8) / (8+5) = (3)(9.8) / 13 = 29.4/13$

$a = 2.26 \text{ m/s}^2$

$T = m_2(g+a) = 5(9.8+2.26) = 5(12.06)$

$T = 60.3 \text{ N}$

M7 · Work, Energy & Power

6 items on the exam — mixes quick concept questions with a couple of computations. Good module for easy points.

Key Ideas

- Work is a SCALAR (even though it's computed from two vectors — force and displacement — via the dot product). No direction to a Joule.
- No work is done unless the object actually moves (displaces) AND there's some component of the applied force along the direction of that displacement. Holding a heavy box still does ZERO work in the physics sense, no matter how tired your arms get.
- If force and displacement are in the same direction, W is positive. If opposite, W is negative. If perpendicular ($\theta=90^\circ$), $W = 0$.
- Kinetic energy (KE) = energy of motion = $\frac{1}{2}mv^2$. Potential energy (PE) = stored energy due to position (like height above the ground).
- The Work-Energy Theorem: the NET work done on an object equals its change in kinetic energy: $W_{\text{net}} = \Delta KE = KE_f - KE_0$.
- Power is about RATE — how quickly work gets done. The same amount of work done faster means more power, not more work.

Core Formulas

$$W = Fd \cos\theta$$

$$W_{\text{net}} = KE_f - KE_0 = \Delta KE$$

$$KE = \frac{1}{2}mv^2$$

$$PE = mgh$$

$$P = W/t = \text{Force} \times \text{velocity}$$

Worked Example — Work at an angle

Given: $F = 60$ N applied at 30° above the direction of motion, object moves 8 m

Required: Work done by this force

$$W = Fd \cos\theta = (60)(8)(\cos 30^\circ) = (480)(0.866)$$

Answer: $W = 415.7$ J

Worked Example — Work-energy theorem

Given: A 2 kg object speeds up from 3 m/s to 9 m/s

Required: Net work done on the object

$$KE_0 = \frac{1}{2}mv_0^2 = \frac{1}{2}(2)(3^2) = \frac{1}{2}(2)(9) = 9 \text{ J}$$

$$KE_f = \frac{1}{2}mv_f^2 = \frac{1}{2}(2)(9^2) = \frac{1}{2}(2)(81) = 81 \text{ J}$$

$$W_{\text{net}} = KE_f - KE_0 = 81 - 9$$

Answer: $W_{\text{net}} = 72$ J

Practice Questions

39

[M7] Is Work a scalar or a vector quantity? What is its SI unit?

Answer: Work is a SCALAR quantity. Its SI unit is the Joule (J).

40

[M7] A person carries a 10 kg bag at constant height while walking 20 m horizontally. How much work does the person do ON the bag (against gravity)?

Answer: Zero. The force (upward, supporting the bag's weight) is perpendicular to the horizontal displacement — $\cos 90^\circ = 0$, so $W = 0$ for that force.

41

[M7] A crate is pushed 5 m across the floor by a 40 N force applied in the exact same direction as the motion. Find the work done.

Answer: $\theta = 0^\circ$ (same direction), so $W = Fd \cos 0^\circ = (40)(5)(1)$

W = 200 J

42

[M7] A 1500 kg car slows from 20 m/s to 5 m/s. Find the net work done on the car (note the sign).

Answer: $KE_0 = \frac{1}{2}(1500)(20^2) = \frac{1}{2}(1500)(400) = 300,000 \text{ J}$

$KE_f = \frac{1}{2}(1500)(5^2) = \frac{1}{2}(1500)(25) = 18,750 \text{ J}$

$W_{\text{net}} = KE_f - KE_0 = 18,750 - 300,000$

$W_{\text{net}} = -281,250 \text{ J}$ (negative, since the car lost kinetic energy)

43

[M7] A motor does 3600 J of work in 12 seconds. Find its power output.

Answer: $P = W/t = 3600/12$

P = 300 W

44

[M7] A machine lifts a 50 kg load at a constant velocity of 0.4 m/s straight up. What power must the machine supply? (Use $P = \text{Force} \times \text{velocity}$, where Force = weight of the load.)

Answer: $F = mg = (50)(9.8) = 490 \text{ N}$

$P = Fv = (490)(0.4)$

P = 196 W

M8 · Impulse & Momentum

3 items on the exam — smallest weight, but the formulas are simple. Easy points if you know the pattern.

Key Ideas

- Momentum (p) = 'mass in motion' = mv . It's a VECTOR — direction matters, so a change in direction (like a ball bouncing back) counts as a big change in momentum even if speed stays the same.
- Impulse (I) equals the CHANGE in momentum: $I = \Delta p = m(v_f - v_0)$. If a ball hits the floor moving down and bounces back up, v_0 and v_f have OPPOSITE signs — don't forget the sign flip.
- Conservation of momentum: in a closed system (no external forces), total momentum before a collision = total momentum after.
- Perfectly ELASTIC collision: both momentum AND kinetic energy are conserved. Perfectly INELASTIC collision: objects stick together and move as one after impact — momentum is conserved, but kinetic energy is NOT (some is lost as heat/sound/deformation).

Core Formulas

$$p = mv$$

$$I = \Delta p = m(v_f - v_0)$$

$$\text{Elastic collision: } m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$$

$$\text{Perfectly inelastic: } m_1v_1 + m_2v_2 = (m_1+m_2)v_f$$

Worked Example — Impulse with a bounce (sign flip)

Given: A 0.2 kg ball hits a wall moving at 12 m/s and bounces straight back at 8 m/s

Required: Impulse delivered to the ball

- Set the initial direction (toward the wall) as positive: $v_0 = +12$ m/s, $v_f = -8$ m/s (bounced BACK, so it's negative)
- $I = m(v_f - v_0) = 0.2(-8 - 12) = 0.2(-20)$

Answer: $I = -4$ kg·m/s (the negative sign shows the impulse acted opposite to the ball's original direction)

Worked Example — Perfectly inelastic collision

Given: Cart A: $m_1 = 3$ kg moving at 6 m/s. Cart B: $m_2 = 2$ kg at rest. They collide and stick together.

Required: Final common velocity v_f

- $m_1v_1 + m_2v_2 = (m_1+m_2)v_f$
- $(3)(6) + (2)(0) = (3+2)v_f$
- $18 = 5v_f \rightarrow v_f = 18/5$

Answer: $v_f = 3.6$ m/s (in the direction Cart A was originally moving)

Practice Questions

45

[M8] A 0.5 kg ball moving at 4 m/s is brought to rest in 0.1 s. Find the impulse delivered to the ball.

Answer: $I = \Delta p = m(v_f - v_0) = 0.5(0 - 4)$

$I = -2$ kg·m/s

46

[M8] In a perfectly inelastic collision, what happens to kinetic energy that is true, but does NOT happen in a perfectly elastic collision?

Answer: Kinetic energy is **NOT** conserved in a perfectly inelastic collision — some KE is lost (converted to heat, sound, or deformation) as the objects stick together. In a perfectly elastic collision, KE **IS** conserved.

47

[M8] A 1200 kg car moving at 15 m/s rear-ends a stationary 1000 kg car, and they lock together (perfectly inelastic). Find their common velocity right after impact.

Answer: $m_1v_1 + m_2v_2 = (m_1+m_2)v_f$
 $(1200)(15) + (1000)(0) = (1200+1000)v_f$
 $18,000 = 2200v_f \rightarrow v_f = 18,000/2200$
 $v_f = 8.18 \text{ m/s}$

M9 · Equilibrium & Torque

3 items on the exam — smallest weight. Focus on the two conditions of equilibrium and the torque formula.

Key Ideas

- 'Equilibrium' means an object is NOT accelerating — it's either completely at rest, or moving at constant velocity. This requires BOTH conditions below to be satisfied.
- 1st condition of equilibrium: $\Sigma F = 0$. In practice this means the x-components of all forces sum to zero AND the y-components sum to zero, checked separately.
- 2nd condition of equilibrium: $\Sigma \text{Torque} = 0$. The sum of clockwise torques must equal the sum of counterclockwise torques.
- Torque depends on THREE things: the force applied, the distance from the pivot/axis (called the lever arm, r), and the ANGLE between the force and the lever arm. Torque is maximized when the force is applied perpendicular to the lever arm ($\theta=90^\circ$), and is ZERO when the force is applied parallel to the lever arm (pushing straight along it, $\theta=0^\circ$) — pushing a door directly toward its hinge doesn't open it.
- A longer lever arm means you need LESS force to produce the same torque — this is why a longer wrench makes it easier to loosen a bolt.

Core Formula

$$\tau = rF \sin\theta$$

Note: θ is the angle between the force vector and the lever arm (r). When force is applied exactly perpendicular to r , $\sin\theta = 1$, and $\tau = rF$ (maximum torque for that force/distance combo).

Worked Example — Torque at an angle

Given: A wrench of length 0.25 m has a 150 N force applied at an angle of 70° to the wrench's length

Required: Torque produced

- $\tau = rF \sin\theta = (0.25)(150)(\sin 70^\circ)$
- $\tau = (37.5)(0.9397)$

Answer: $\tau = 35.24 \text{ N}\cdot\text{m}$

Worked Example — Equilibrium — balancing torques

Given: A 4 m uniform seesaw is pivoted at its center. A 30 kg child sits 1.5 m from the pivot on the left side. Where must a 45 kg child sit on the right side to balance it?

Required: Distance r_2 from the pivot for the 45 kg child

- For balance: clockwise torque = counterclockwise torque $\rightarrow (m_1g)(r_1) = (m_2g)(r_2)$
- The g cancels from both sides: $m_1r_1 = m_2r_2 \rightarrow (30)(1.5) = (45)(r_2)$
- $45 = 45r_2 \rightarrow r_2 = 45/45$

Answer: $r_2 = 1 \text{ m}$ from the pivot

Practice Questions

48

[M9] What are the TWO conditions that must both be true for an object to be in complete equilibrium?

Answer: (1) $\Sigma F = 0$ (net force is zero — both x and y components sum to zero), and (2) $\Sigma \text{Torque} = 0$ (sum of clockwise torques equals sum of counterclockwise torques).

49

[M9] A force of 80 N is applied exactly perpendicular to a lever arm that is 0.4 m long. Find the torque produced.

Answer: $\theta = 90^\circ$, so $\sin\theta = 1$. $\tau = rF \sin\theta = (0.4)(80)(1)$

$\tau = 32 \text{ N}\cdot\text{m}$

50

[M9] If you apply a force directly along the length of a wrench (pulling it straight out, parallel to the handle) instead of perpendicular to it, what torque do you produce, and why?

Answer: Zero torque. Since $\theta = 0^\circ$ between the force and the lever arm, $\sin\theta = 0$, so $\tau = rF \sin\theta = 0$ no matter how large the force is.

Final Countdown Checklist

Use this the night before / morning of the exam.

Can you do each of these without looking at your notes?

- Compute the magnitude and angle of inclination of a 2D vector
- Find a dot product and a cross product from components
- Pick the correct kinematics equation given what's provided vs. what's asked for
- Solve a free-fall problem where the object is thrown upward (remember $v_f = 0$ at the peak)
- Break projectile motion into horizontal (constant v) and vertical (free-fall) components
- Compute centripetal acceleration and force from v and r
- Draw a free-body diagram and correctly identify which forces are present (W.A.N.T. Food)
- Set up $\Sigma F_x = \max$ and $\Sigma F_y = \max$ separately for an object on an incline
- Compute acceleration and tension for a pulley system (block + hanging mass, or Atwood machine)
- Apply the work-energy theorem to find net work from a change in KE
- Compute impulse, remembering to flip the sign when a velocity reverses direction (bounce)
- Distinguish elastic vs. perfectly inelastic collisions and know which quantities are conserved in each
- Apply both conditions of equilibrium, and compute torque using $\tau = rF \sin\theta$

Common Sign & Unit Traps

- g is NEGATIVE (-9.8 m/s^2) when 'up' is your positive direction — this is the convention used throughout M3 and M4 in your modules.
- Work, dot product, and speed are SCALARS (no direction/sign to worry about in the same way). Displacement, velocity, momentum, and torque are VECTORS — direction/sign matters.
- Convert angles to RADIANS before using $s = r\theta$, but keep angles in DEGREES for $\sin/\cos/\tan$ on your calculator (check your calculator is in degree mode, as your modules remind you).
- μ_s (static) is always $\geq \mu_k$ (kinetic) for the same pair of surfaces — it takes more force to start something moving than to keep it moving.
- Double check units are consistent (e.g., convert grams to kg, minutes to seconds) before plugging into a formula — this is a common source of careless errors.

You've got this. Get there early, breathe, and work through the easy points first — Good luck on GED0081!